Referee Report: “Resolving Coordination Frictions in Green Labor Transitions” by Shisham Adhikari

Summary

This paper studies how government policies can promote labor reallocation to targeted sectors of the economy by alleviating search-and-matching frictions in the context of segmented labor markets. It builds a quantitative model in which workers face a higher cost of entering the targeted sector and firms need to pay a cost when posting vacancies. Together, these frictions can keep the targeted sector from expanding, even if there are positive externalities associated with employment in that sector. The paper’s main application of interest is the green transition, which is forecast to require the employment share in green energy and mobility in the US from 2% to 14% between 2022 and 2030. Calibrating the model to the state of the US labor market in 2022, the paper solves for the combination of worker and firm subsidies that yield a green employment share of up to 14% while minimizing unemployment and fiscal costs, while maximizing welfare. The key result is that combining worker and firm subsidies yields better outcomes, with worker subsidies being especially important in the early and late phases of the transition.

Assessment and Main Points

I discussed this paper at the EER-Banxico Conference on the Macro and Financial Aspects of Climate change. I find this submitted version to be highly responsive to the comments I raised at that time. I especially like that this new version makes the lessons of the paper broadly applicable to situations in which externalities call for a big push to reallocate labor towards some sector of the economy. I also like how the paper motivates the policy intervention with an externality from the start, and its emphasis on the benefits of using multiple policy levers at the same time.

Altogether, the paper makes an important contribution by showing how labor market frictions can prevent some sectors in the economy from expanding. Economists and policymakers often fixate on capital markets and other capital frictions as the main barriers to reallocation. This paper challenges that view and goes even further by showing that a two-pronged policy subsidizing both workers and firms can be more effective than a policy limited to one instrument.

After reading the new version, I have two essential comments for the paper and a number of smaller comments and suggestions.

1. **Transition dynamics.** I am satisfied with the way the paper models the green transition; namely, as a sequence of steady states in which the policy mix gradually raises the share of employment in the green sector to 14%. I agree that this fits the reality of gradual policy shifts, and that workers and firms are unlikely to have perfect foresight over how taxes/subsidies will evolve even within a decade. That said, I think modeling a sequence of steady states will seem much more plausible if the paper can show that the economy transitions to a new steady state *quickly* after unexpected shifts in policy (like MIT shocks). I believe the DMP-style model used in this paper will yield a quick transition because there are no forces (like incentives for consumption smoothing, or adjustment costs) dampening the short-term reaction of the labor market. However, it would be great to confirm that hypothesis quantitatively.

2. **Identification.** I appreciate the paragraph on page 22 that explicitly maps parameters in the calibration to targeted moments. But the paper is missing a clear, intuitive explanation for that mapping. This is important because in a calibration like this one, it is often the case that all parameters affect all targeted moments to some degree. *Why* does the separation rate λ shift the unemployment rate more than the vacancy posting cost c, for example? This discussion might be easier if the paper plots how key target moments vary when changing the value of key parameters.

Smaller Comments and Suggestions

- In the top paragraph of page 21, I think the paper should clarify that the hiring likelihood ratio for workers with green skills should be "29% higher than for other workers" to make the explanation consistent with the 1.29 value in Table 2.

- Also on page 21, I don't understand what the paper means when it says "using 2022 generation is conservative given the near-term pipeline." I also don't understand what this means: "the higher benchmark provides a fiscal stress test."

- Why does the calibration use a 2023 unemployment rate of 3.5%, while the paper claims the calibration is based on the 2022 labor market? Is this a typo?
- On page 24, the paragraph under Table 3 refers to three policy paths. I don't think these are paths yet, but rather three policy mixes that yield the 14% employment share in the green sector.
- I found the writing in Section 4.3.1 a bit convoluted, but I am happy with the quantitative exercise described there.
- In Figure 7, why aren't all the bars the same width?
- I think the biggest takeaway from Figure 7 is the contrast between big swings in the optimal policy mix and the near-constant size of the intervention needed to generate higher green employment shares. The hump shape in the total fiscal cost seems like a second-order result by comparison.
- In terms of framing, I think the size of the policy interventions in the paper are likely an upper bound. Technical change will likely raise the productivity/wages of the green sector, and young generations of workers will likely face lower costs of entering the green sector. Both of these trends should alleviate the frictions in the paper over time, although not fast enough to yield a 14% green employment share by 2030.
- I found Figure D.8 in the appendix to be very illustrative. I wonder if it might deserve to be in the main text.